

JOSEPH JIN

josephjin@berkeley.edu | josephjin.dev | linkedin.com/in/joseph-jin | github.com/AnimatorJoe

Education

University of California, Berkeley

Class of 2024

B.S. Electrical Engineering and Computer Sciences

GPA: 4.0/4.0

Relevant Coursework:

Machine Learning, Artificial Intelligence, Advanced Algorithms, Data Structures, Optimization Models, Discrete Math and Probability, Computer Architecture, Probability for Data Science, Structure of Computer Programs, Circuit Design and Theory I/II, Physics for Engineers

Experience

Software Engineering Intern, Autodesk

May 2021 - August 2021

- Upgraded tamper-detection software in C++ to report pirating attempts on Autodesk's 100+ products used by ~4 million customers by utilizing native MacOS and Windows SSL certificate verification libraries and writing unit and integration tests for common pirating techniques
- Programmed and deployed internal product for monitoring errors on customer-facing servers (capturing 400+ issues during internship) with Javascript and Jenkins

Undergraduate Researcher, Berkeley Artificial Intelligence Research Lab

February 2022 - Present

- Worked with Prof. Trevor Darrel and PhD candidate Zhuang Liu to research the impact of tuning regularization parameters (ex: weight decay, drop path, data augmentation) of neural networks during training time on the model's accuracy and convergence speed
- Designed and conducted regularization tuning experiments on convolutional neural networks by adapting models, such as ResNets and ConvNeXts, for trials using PyTorch, timm, and Pandas

Embedded Software Intern, Silicon Labs

June 2019 - August 2019

- Programmed Embedded C software on Silicon Labs microcontrollers used by enterprise customers as design reference
- Implemented bluetooth support for 32-bit microcontrollers after discovering missing compatibility outside 8-bit systems
- Wrote and patched reference code for low-energy peripherals on 32-bit systems

Projects

Data Consultant, Tribe Capital

February 2022 - May 2022

- Developed end-to-end ML labeling, training and prediction pipeline for Tribe Capital (venture capital firm) to identify prospective startups as part of data consulting project under Berkeley Student Association for Applied Statistics
- Trained ensemble learning models and searched hyperparameters using Pandas, Sklearn, and NumPy, achieving a prediction accuracy of 90% based on past companies outside training set

Visualize - Text-to-Speech Accessibility Tool for the Blind

- Led team of four engineers to design, develop, and deploy Visualize, a mobile application which helps the blind "see" by using text-to-speech to inform users what their phone camera is pointing at; received 9,000+ downloads
- Achieved object detection in images using Apple's Core ML framework and Microsoft's Computer Vision API on top of Swift

Co-Founder, Stroll

April 2021 - December 2021

- Led team of five engineers to develop and deploy algorithm, backend, and frontend codebase for safety based GPS app Stroll, enrolled in SkyDeck Batch 13 incubator program (3% acceptance rate)
- Designed and developed safety based GPS routing algorithm for the city of Berkeley using OpenStreetMap, Pandas, and NumPy
- Architected and implemented AWS SAM backend and React Native frontend in Javascript

Skills

Languages: Python, Java, C, Golang, Swift, Javascript, R, Matlab

Frameworks/Tools: PyTorch, Scikit-learn, NumPy, Pandas, CVXPY, Git, AWS, MySQL, MongoDB

Frontend/Backend: React.js, React Native, Vue.js, Node.js, Apple UIKit, Selenium

- **Eta Kappa Nu Honors Society** - top 20% of Berkeley EECS undergraduate students
- **Gold Division, USACO** - top 6% of competitors in the United States
- **Finalist, GitHub GameOff** - ranked top 3% out of all competitors
- **1st Place Team, TACC Coding Competition** - invitational coding contest between 50 teams at the University of Texas, Austin
- **2nd Place, Best of Texas** - state wide programming contest